Practical Questions

1. Write a program to save a trained machine learning model using joblib.dump().

In [1]:
```
from sklearn.datasets import load_iris
from sklearn.ensemble import RandomForestClassifier
import joblib

# Load dataset
iris = load_iris()

X = iris.data
y = iris.target

# Train model
model = RandomForestClassifier()

model.fit(X, y)

# Save model
joblib.dump(model,  "iris_model.pkl")

print("Model saved successfully")
```

```
Model saved successfully
```

2. Write a Python program to load a saved model using joblib.load() and make predictions.

In [2]:
```
import joblib

# Load saved model
model = joblib.load("iris_model.pkl")

# Sample input
sample_data = [[5.1, 3.5, 1.4, 0.2]]

# Prediction
prediction = model.predict(sample_data)

print("Predicted Class:", prediction)
```

```
Predicted Class: [0]
```

3. Write a Tkinter program to create a GUI window with the title “Iris Flower Classification”.

In [3]:
```
from tkinter import *
# Create window
root = Tk()

# Window title
root.title("Iris Flower Classification")

# Window size
root.geometry("400x300")

# Run application
root.mainloop()
```

4. Write a Tkinter program to create labels and entry boxes for Sepal Length and Sepal Width.

In [4]:
```
from tkinter import *
root = Tk()
```

```
root.title("Iris Flower Classification")

# Sepal Length
label1 = Label(root, text="Sepal Length")
label1.pack()

entry1 = Entry(root)
entry1.pack()

# Sepal Width
label2 = Label(root, text="Sepal Width")
label2.pack()

entry2 = Entry(root)
entry2.pack()

root.mainloop()
```

5. Write a Python program to create a button in Tkinter that displays a popup message when clicked.

In [5]:
```
from tkinter import *
from tkinter import messagebox
root = Tk()
root.title("Popup  Example")

# Function
def show_message():
    messagebox.showinfo("Message",  "Button  Clicked")

# Button
button = Button(root, text="Click Me", command=show_message)
button.pack(pady=20)
root.mainloop()
```

6. Write a Python code snippet to handle invalid numeric input using try-except ValueError.

In [6]:
```
try:
    number = float(input("Enter a number: "))
    print("You entered:", number)
except ValueError:
    print("Invalid numeric input")
```

```
You entered: 4.0
```

7. Write a program to check whether the file iris_model.pkl exists using the os module.

In [7]:
```
import os

if os.path.exists("iris_model.pkl"):
    print("File exists")

else:
    print("File does not exist")
```

```
File exists
```

8. Write a Python program to create four input fields using Tkinter Entry widgets.

In [8]:
```
from tkinter import *
root = Tk()
root.title("Iris Input Form")
# Sepal Length
Label(root, text="Sepal Length").pack()
```

```
entry1 = Entry(root)
entry1.pack()

# Sepal Width
Label(root, text="Sepal Width").pack()
entry2 = Entry(root)
entry2.pack()

# Petal Length
Label(root, text="Petal Length").pack()
entry3 = Entry(root)
entry3.pack()

# Petal Width
Label(root, text="Petal Width").pack()
entry4 = Entry(root)
entry4.pack()

root.mainloop()
```

9. Write code to predict the flower category using model.predict() for user-entered values.

In [9]:
```
import joblib

# Load model
model = joblib.load("iris_model.pkl")

# User values
sepal_length = 5.1
sepal_width = 3.5
petal_length = 1.4
petal_width = 0.2

# Prediction
prediction = model.predict([
    [sepal_length, sepal_width, petal_length, petal_width]
])
print("Prediction:", prediction)
```

```
Prediction: [0]
```

10. Write a Python program to display the predicted flower name using messagebox.showinfo().

In [10]:
```
from tkinter import *
from tkinter import messagebox
root = Tk()
flower_name = "Setosa"
messagebox.showinfo("Prediction",f"Predicted Flower: {flower_name}")
root.mainloop()
```

11. Write a program to create a fixed-size Tkinter window using geometry() and resizable().

In [11]:
```
from tkinter import *
root = Tk()
root.title("Fixed Size Window")

# Window size
root.geometry("400x300")

# Disable resizing
root.resizable(False, False)

root.mainloop()
```

12. Write a Python program that loads the Iris dataset and prints the target names (setosa, versicolor, virginica).

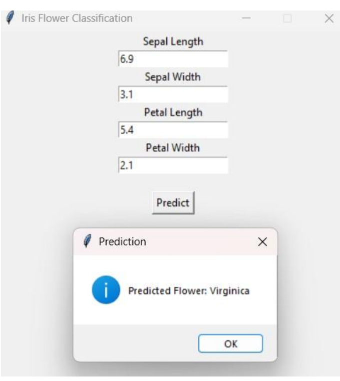
In [12]:
```
from sklearn.datasets import load_iris

# Load dataset
iris = load_iris()

# Print target names
print("Target  Names:")

print(iris.target_names)
```

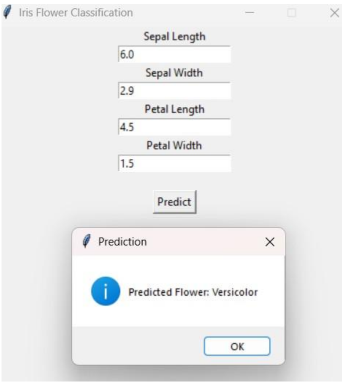
```
Target Names:
['setosa' 'versicolor' 'virginica']
```

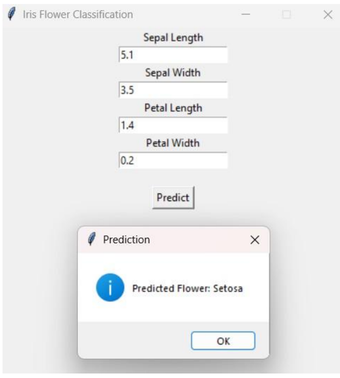
Complete GUI Window Run

In [13]:
```
from tkinter import *
from tkinter import messagebox
import joblib

# Load trained model
model = joblib.load("iris_model.pkl")

# Create window
root = Tk()

root.title("Iris Flower Classification")
root.geometry("400x400")
root.resizable(False,  False)

# Labels and Entry Fields
Label(root, text="Sepal Length").pack()

entry1 = Entry(root)
entry1.pack()
Label(root, text="Sepal Width").pack()

entry2 = Entry(root)
entry2.pack()
Label(root, text="Petal Length").pack()

entry3 = Entry(root)
entry3.pack()
Label(root, text="Petal Width").pack()
entry4 = Entry(root)
entry4.pack()
# Prediction Function
def predict_flower():
    try:
        sl = float(entry1.get())
        sw = float(entry2.get())
        pl = float(entry3.get())
        pw = float(entry4.get())
        prediction = model.predict([
            [sl, sw, pl, pw]
        ])
        flowers = [
            "Setosa",
            "Versicolor",
            "Virginica"
        ]
        result = flowers[prediction[0]]
        messagebox.showinfo(
            "Prediction",
```

```
            f"Predicted Flower: {result}"
        )
    except ValueError:
        messagebox.showerror(
            "Error",
            "Please enter valid numeric values"
        )

# Predict Button
button = Button(
    root,
    text="Predict",
    command=predict_flower
)
button.pack(pady=20)

# Run application
root.mainloop()
```







Submitted By:

- Name: Amulya Shrivastava
- Regd No: 2341019456
- Section: 23412F3
- Serial No: 59
- Semester: 6th
- Branch: B.Tech CSE Core